



Advanced SOC Operations and Threat Analysis Report

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Date:

22-05-2026

INTRODUCTION:

This practical task was performed to understand advanced SOC operations including log correlation, threat intelligence integration, incident escalation workflows, alert triage, evidence preservation, and full SOC workflow simulation activities. Different monitoring, investigation, escalation, and reporting procedures were documented to understand how SOC analysts detect, investigate, prioritize, and respond to cybersecurity incidents in real-world environments.

OBJECTIVES:

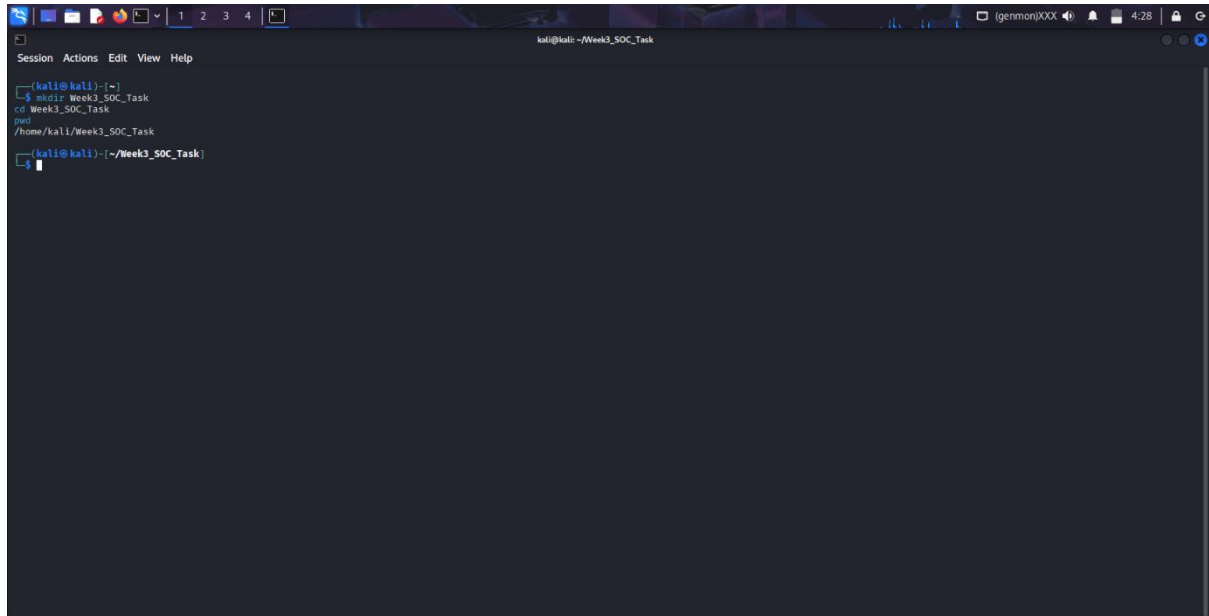
- To understand advanced log analysis techniques
- To learn threat intelligence integration concepts
- To practice incident escalation workflow
- To understand alert triage and IOC validation
- To learn evidence preservation procedures
- To simulate a complete SOC workflow investigation

TOOLS USED:

- Kali Linux
- Microsoft Word
- Elastic Security
- Wazuh
- AlienVault OTX
- VirusTotal
- Metasploit Framework
- GitHub



Environment Setup:



```
kali@kali:~/Week3_SOC_Task
Session Actions Edit View Help
[kali@kali]~$
[kali@kali]~$ mkdir Week3_SOC_Task
[kali@kali]~$ cd Week3_SOC_Task
[kali@kali]~/Week3_SOC_Task$ pwd
/home/kali/Week3_SOC_Task
[kali@kali]~/Week3_SOC_Task$
```

Figure 1: SOC Environment Setup in Kali Linux

Advanced Log Correlation Analysis:

Timestamp	Event ID	Source IP	Destination IP	Severity	Status	Analyst Notes
2026-05-20 12:00:00	4625	192.168.1.100	8.8.8.8	High	Investigating	Suspicious DNS request
2026-05-20 12:05:00	5156	192.168.1.100	45.33.32.156	Medium	Open	Unusual Outbound traffic
2026-05-20 12:10:00	4104	192.168.1.101	192.168.1.1	High	Escalated	Suspicious PowerShell activity

Explanation:

The advanced log correlation process was performed to understand how SOC analysts identify suspicious activities by analyzing logs collected from multiple security sources. Different events such as failed authentication attempts, unusual outbound traffic connections, and suspicious PowerShell execution activities were correlated using timestamps, IP addresses, and event identifiers. This activity helped in



understanding how security teams detect attack patterns, investigate suspicious behavior, and reduce false positives during monitoring operations in SOC environments.

Observation:

The correlated logs indicated suspicious authentication attempts and abnormal outbound traffic activity originating from internal systems. Event correlation improved visibility into suspicious behavior and supported better incident investigation workflow during SOC monitoring activities.

Learning Outcome:

This activity improved understanding of log correlation techniques, event analysis, and suspicious activity monitoring procedures commonly followed by SOC analysts during incident detection and investigation processes.

Anomaly Detection Analysis:

Rule Name	Detection Condition	Source IP	Severity	Status	Analyst Notes
High Data Transfer	bytes_out > 1MB in 1m	192.168.1.101	High	Triggered	Unusual outbound traffic detected
Suspicious Login Time	Login outside working hours	192.168.1.102	Medium	Open	Abnormal authentication timing
Excessive DNS Requests	More than 50 requests in 1 minute	192.168.1.103	Medium	Investigating	Possible DNS Tunnelling activity

Explanation:

The anomaly detection activity was performed to understand how SOC analysts identify unusual system behavior using rule-based monitoring techniques. Different abnormal activities such as high-volume outbound



traffic, suspicious login timings, and excessive DNS requests were analyzed during monitoring operations. This process helped in understanding how security monitoring systems detect potential threats and generate alerts for further investigation during SOC activities.

Observation:

The monitoring process identified abnormal outbound traffic patterns and suspicious authentication activities originating from internal systems. The generated alerts provided better visibility into unusual system behavior and supported additional investigation procedures during SOC monitoring operations.

Learning Outcome:

This activity improved understanding of anomaly detection techniques, suspicious activity monitoring, and rule-based alert generation processes commonly used in SOC environments for threat detection and investigation procedures.

Log Enrichment Analysis:

IP Address	Country	Risk Level	Threat Source	Status	Analyst Notes
45.33.32.156	United States	Medium	AlienVault OTX	Investigating	Suspicious outbound connection
185.220.101.1	Germany	High	VirusTotal	Escalated	Linked to malicious activity
192.168.1.101	Internal Network	Medium	Internal Monitoring	Open	Suspicious PowerShell execution

Explanation:

The log enrichment process was performed to improve security analysis by adding contextual information to suspicious events and IP addresses identified during monitoring activities. Details such as geolocation, risk level, and threat intelligence references were added to the collected logs



to provide better visibility into suspicious network activity. This activity helped in understanding how enriched security data supports faster and more accurate incident investigation procedures in SOC environments.

Observation:

The enriched logs provided additional visibility into suspicious network traffic and potentially malicious IP addresses. Contextual information such as geolocation and threat intelligence references improved the efficiency of alert investigation and supported better decision-making during monitoring activities.

Learning Outcome:

This activity improved understanding of log enrichment techniques, contextual threat analysis, and enhanced security monitoring processes commonly used by SOC analysts during incident investigation and threat detection operations.

Threat Intelligence Integration:

Feed Source	IOC Type	IOC Value	Risk Level	Status	Analyst Notes
AlienVault OTX	Malicious IP	192.168.1.100	High	Detected	Linked to suspicious outbound activity
VirusTotal	File Hash	A45F7C9D2E	Medium	Investigating	Suspicious file reputation detected
Internal Threat Feed	Domain	Suspicious-login.com	High	Escalated	Potential phishing domain identified

Explanation:

Threat intelligence integration activities were performed to understand how SOC analysts use external and internal threat feeds to identify suspicious indicators during monitoring operations. Different indicators of compromise such as malicious IP addresses, suspicious file hashes, and



phishing domains were analyzed using threat intelligence sources including AlienVault OTX and VirusTotal. This activity helped improve understanding of alert validation, IOC analysis, and threat detection procedures commonly followed in SOC environments.

Observation:

The integrated threat intelligence feeds provided additional visibility into suspicious network indicators and potentially malicious activity. IOC analysis improved the efficiency of alert investigation and helped identify high-risk indicators requiring escalation during monitoring operations.

Learning Outcome:

This activity improved understanding of threat intelligence integration, IOC validation, and enriched alert analysis processes commonly used by SOC analysts during threat monitoring and incident investigation activities.

Alert Enrichment Analysis:

Alert ID	IP Address	Reputation	Threat Source	Notes
003	192.168.1.100	Malicious	AlienVault OTX	Linked to suspicious outbound traffic
004	185.220.101.1	High Risk	VirusTotal	Associated with malicious activity
005	192.168.1.101	Medium Risk	Internal Monitoring	Suspicious PowerShell execution

Explanation:

Alert enrichment activities were performed to add contextual threat intelligence information to suspicious security alerts identified during monitoring operations. Reputation details obtained from threat intelligence platforms helped provide better visibility into malicious indicators and supported more accurate alert investigation procedures within the SOC workflow.



Observation:

The enriched alerts provided additional context regarding suspicious IP addresses and threat indicators. Threat intelligence data improved alert prioritization and supported better investigation decisions during monitoring and incident response activities.

Learning Outcome:

This activity improved understanding of alert enrichment techniques, threat reputation analysis, and intelligence-driven SOC monitoring procedures used during cybersecurity investigations.

Threat Hunting Activity:

MITRE ATT&CK Technique:

T1078 – Valid Accounts

Hunting Query:

user.name != "system"

Findings:

Suspicious account activity and abnormal login behavior were identified during log monitoring operations. Multiple authentication attempts were observed outside standard working hours, requiring additional investigation.

Explanation:

Threat hunting activities were performed to understand how SOC analysts proactively search for suspicious behavior within security logs and monitored systems. The investigation focused on identifying abnormal account usage patterns related to MITRE ATT&CK Technique T1078 (Valid Accounts). This activity helped improve understanding of proactive threat detection and behavioral analysis procedures used in SOC operations.



Observation:

The threat hunting process identified unusual account activity and suspicious authentication behavior within the monitored environment. Behavioral analysis techniques improved visibility into potentially unauthorized access attempts during monitoring activities.

Learning Outcome:

This activity improved understanding of proactive threat hunting techniques, account behavior analysis, and MITRE ATT&CK-based investigation procedures commonly followed during SOC monitoring operations.

Incident Escalation Workflow:

Alert ID	Severity	Assigned Tier	Escalation Reason	Status	Analyst Notes
INC-101	High	Tier 2	Unauthorized access attempt detected	Escalated	Suspicious login activity observed
INC-102	Medium	Tier 1	Abnormal outbound traffic detected	Investigating	Requires additional monitoring
INC-103	High	Tier 2	Multiple failed Authentication attempts	Escalated	Potential brute-force activity

Explanation:

The incident escalation workflow was performed to understand how security incidents are transferred between different SOC analyst tiers based on severity and complexity. High-priority incidents such as unauthorized access attempts, abnormal network activity, and repeated authentication failures were escalated to Tier 2 analysts for advanced investigation and response activities. This process helped improve understanding of structured incident handling procedures followed in SOC environments.



Observation:

The escalation workflow improved visibility into incident prioritization and analyst responsibilities during monitoring operations. High-severity alerts required immediate escalation to higher analyst tiers for advanced investigation and response procedures.

Learning Outcome:

This activity improved understanding of SOC escalation procedures, incident prioritization, and structured analyst workflows commonly followed during cybersecurity monitoring and incident response activities.

Situation Report (SITREP):

Incident Title:

Unauthorized Access on Server-Y

Incident Summary:

Suspicious authentication activity was detected on Server-Y at 2026-05-20 13:00:00. Multiple failed login attempts were observed from source IP address 192.168.1.200. The detected activity was mapped to MITRE ATT&CK Technique T1078 – Valid Accounts.

Actions Taken:

- Isolated the affected system
- Escalated incident to Tier 2 analysts
- Monitored suspicious network traffic
- Documented investigation findings

Explanation:

The Situation Report (SITREP) was prepared to document the incident summary, affected systems, suspicious activities, and response actions performed during the escalation process. Structured incident communication helps SOC teams coordinate investigation activities and



maintain proper documentation throughout the incident response lifecycle.

Observation:

The SITREP process improved communication efficiency between SOC analysts and incident response teams. Structured reporting provided better visibility into incident status, response actions, and escalation procedures.

Learning Outcome:

This activity improved understanding of structured incident communication, escalation documentation, and SOC reporting procedures commonly followed during cybersecurity investigations.

Workflow Automation:

Automation Rule:

If Alert Severity = High
→ Automatically Assign Incident to Tier 2 Analyst

Automation Status:

Successfully Configured

Response Action:

Automatic ticket escalation and analyst notification initiated during alert generation.

Explanation:

Workflow automation concepts were studied to understand how SOAR platforms automatically assign and manage high-priority alerts during incident response operations. Automated escalation procedures improved response efficiency and reduced manual workload during monitoring and investigation activities within SOC environments.



Observation:

The workflow automation process improved alert handling efficiency and reduced manual response delays during incident escalation procedures. Automated analyst assignment supported faster response coordination during monitoring activities.

Learning Outcome:

This activity improved understanding of SOAR workflow automation, automated escalation procedures, and incident response coordination processes commonly implemented in SOC operations.

Alert Triage with Threat Intelligence:

Alert ID	Alert Description	Source IP	Priority	Status	Analyst Notes
ALT-201	Suspicious PowerShell Execution	192.168.1.101	High	Open	Potential malicious script activity detected
ALT-202	Abnormal Authentication Attempt	192.168.1.102	Medium	Investigating	Suspicious login behavior observed
ALT-203	Excessive DNS Requests	192.168.1.103	Medium	Escalated	Possible DNS tunnelling activity

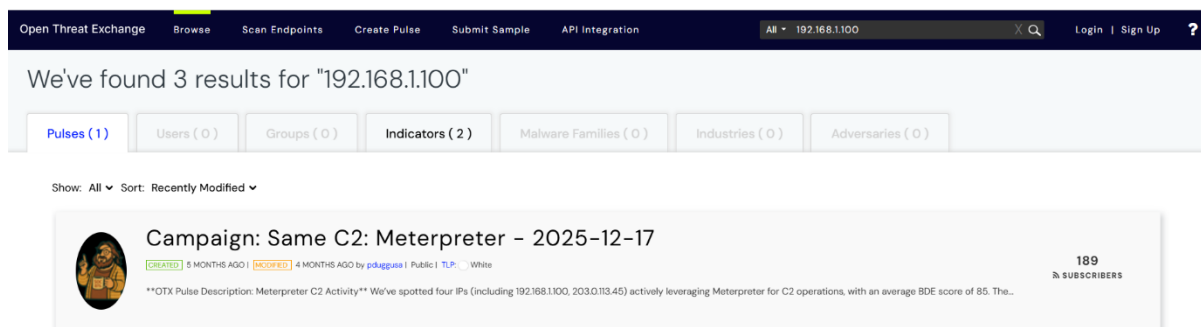


Figure 2: Threat Intelligence Analysis using AlienVault OTX

Explanation:

The alert triage process was performed to analyze and prioritize suspicious security alerts identified during monitoring operations. Different alerts such as suspicious PowerShell execution, abnormal authentication attempts, and excessive DNS requests were reviewed using threat intelligence concepts and IOC analysis techniques. This activity helped improve understanding of alert prioritization, incident classification, and investigation workflows commonly followed in SOC environments.

Observation:

The alert triage workflow improved visibility into suspicious activities and helped prioritize high-risk alerts for investigation. Threat intelligence data and IOC validation techniques supported more accurate incident classification during monitoring activities.



Learning Outcome:

This activity improved understanding of alert triage procedures, incident prioritization, IOC validation, and intelligence-driven investigation workflows commonly used in SOC operations.

IOC Validation:

Validated Indicator:

185.220.101.1

Threat Intelligence Sources:

- VirusTotal
- AlienVault OTX
- Internal Monitoring System

Validation Result:

The IP address was associated with suspicious activity indicators and abnormal outbound communication patterns requiring additional investigation.

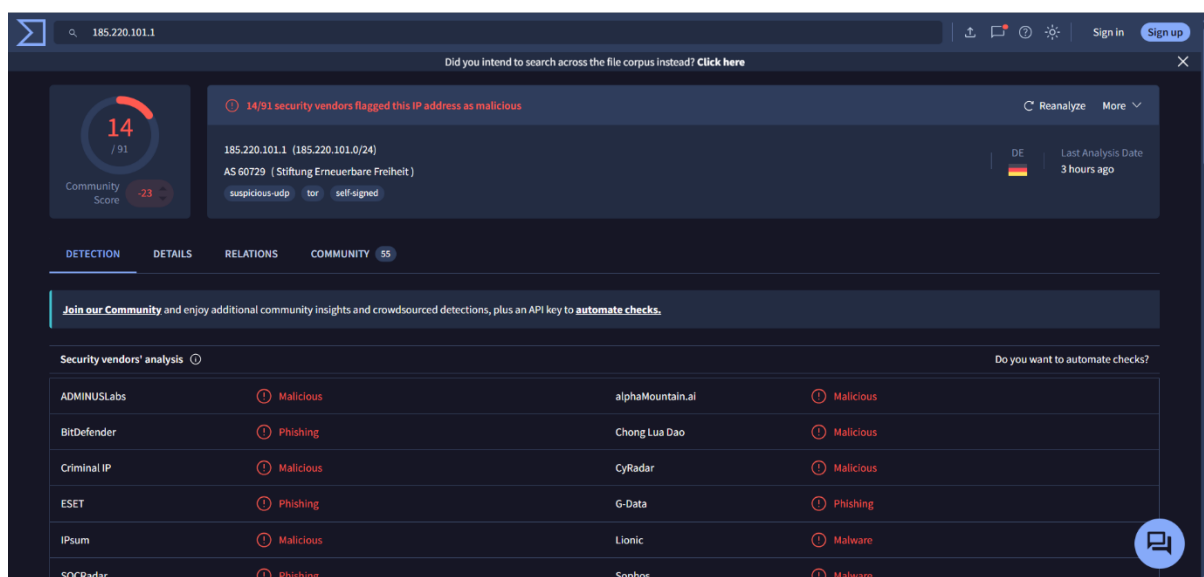


Figure 3: IOC Validation using VirusTotal



Explanation:

IOC validation activities were performed to verify suspicious indicators using threat intelligence platforms such as VirusTotal and AlienVault OTX. Different IP addresses and suspicious indicators were analyzed to provide additional context for alert investigation and improve decision-making during SOC monitoring operations.

Observation:

The IOC validation process provided additional visibility into suspicious indicators and improved the accuracy of alert investigation activities. Threat intelligence references supported better prioritization of high-risk incidents during monitoring operations.

Learning Outcome:

This activity improved understanding of IOC validation techniques, threat intelligence usage, and investigation support procedures commonly followed during cybersecurity monitoring and incident response activities.

Evidence Preservation and Analysis:

Volatile Data Collection:

Command Used:

```
SELECT * FROM netstat
```

Collected Information:

- Active network connections
- Established sessions
- Listening ports
- Running communication processes

Collection Status:

Successful



```
Session Actions Edit View Help
(kali@kali)~$ ps aux
USER      PID CPU %MEM    VIRT   RSS TTY      STAT START   TIME COMMAND
root         1  0.2  0.3 23828 14088 ?        Ss   06:52   0:01 /sbin/init splash
root         2  0.0  0.0      0     0 ?        S    06:52   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    06:52   0:00 [pool_workqueue_release]
root        14  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/R-kvfree_rcu_reclaim]
root         5  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/R-rcu_gp]
root         6  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/R-sync_wq]
root         7  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/R-slab_flushwq]
root         8  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/R-netns]
root        10  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/0:0H-events_highpri]
root        12  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/u128:0-ipv6_addrconf]
root        13  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/R-mm_percpu_wq]
root        14  0.0  0.0      0     0 ?        I    06:52   0:00 [rcu_tasks_kthreadd]
root        15  0.0  0.0      0     0 ?        I    06:52   0:00 [rcu_tasks_rude_kthreadd]
root        16  0.0  0.0      0     0 ?        I    06:52   0:00 [rcu_tasks_trace_kthreadd]
root        17  0.0  0.0      0     0 ?        S    06:52   0:00 [ksoftirqd/0]
root        18  0.0  0.0      0     0 ?        I    06:52   0:00 [rcu_preempt]
root        19  0.0  0.0      0     0 ?        S    06:52   0:00 [rcu_exp_work_gp_kthreadd_worker/1]
root        20  0.0  0.0      0     0 ?        S    06:52   0:00 [rcu_exp_gp_kthreadd_worker]
root        21  0.0  0.0      0     0 ?        S    06:52   0:00 [migration/0]
root        22  0.0  0.0      0     0 ?        S    06:52   0:00 [idle_inject/0]
root        23  0.0  0.0      0     0 ?        S    06:52   0:00 [cpuhp/0]
root        24  0.0  0.0      0     0 ?        S    06:52   0:00 [cpuhp/1]
root        25  0.0  0.0      0     0 ?        S    06:52   0:00 [idle_inject/1]
root        26  0.0  0.0      0     0 ?        S    06:52   0:00 [migration/1]
root        27  0.0  0.0      0     0 ?        S    06:52   0:00 [ksoftirqd/1]
root        28  0.0  0.0      0     0 ?        I    06:52   0:00 [kworker/1:0-mm_percpu_wq]
root        29  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/1:0H-events_highpri]
root        30  0.0  0.0      0     0 ?        S    06:52   0:00 [cpuhp/2]
root        31  0.0  0.0      0     0 ?        S    06:52   0:00 [idle_inject/2]
root        32  0.0  0.0      0     0 ?        S    06:52   0:00 [migration/2]
root        33  0.0  0.0      0     0 ?        S    06:52   0:00 [ksoftirqd/2]
root        35  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/2:0H-events_highpri]
root        36  0.0  0.0      0     0 ?        S    06:52   0:00 [cpuhp/3]
root        37  0.0  0.0      0     0 ?        I    06:52   0:00 [idle_inject/3]
root        38  0.0  0.0      0     0 ?        S    06:52   0:00 [migration/3]
root        39  0.0  0.0      0     0 ?        S    06:52   0:00 [ksoftirqd/3]
root        40  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/3:0-cgroup_destroy]
root        41  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/3:0H-events_highpri]
root        42  0.3  0.0      0     0 ?        I    06:52   0:01 [kworker/u129:0-events_unbound]
root        43  0.0  0.0      0     0 ?        I    06:52   0:00 [kworker/u130:0-events_unbound]
root        47  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/u129:3-kvfree_rcu_reclaim]
root        48  0.0  0.0      0     0 ?        S    06:52   0:00 [kdevtmpfs]
root        49  0.0  0.0      0     0 ?        Ic   06:52   0:00 [kworker/R-inet_frag_wq]
root        50  0.0  0.0      0     0 ?        S    06:52   0:00 [kauditd]
```

Figure 4: Active Network Connection Monitoring using netstat

```
Session Actions Edit View Help
(kali@kali)~$ netstat -an
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:15601           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:119380          0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:119300          0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:9200            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:119200          127.0.0.1:43174         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:59714         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:46328         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:43182         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:40392         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:41002         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:40408         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:46344         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:46390         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:46312         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:43170         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:46342         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:41006         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:59730         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:59784         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:40396         TIME_WAIT
tcp        0      0 0.0.0.0:119200          127.0.0.1:59732         TIME_WAIT
udp        0      0 192.168.112.128:68     192.168.112.254:67     ESTABLISHED
raw        0      0 0.0.0.0:                0.0.0.0:*               7

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State      I-Node   Path
unix    3      [ ]       STREAM    CONNECTED 12882
unix    3      [ ]       STREAM    CONNECTED 11081
unix    3      [ ]       STREAM    CONNECTED 11814  /run/user/1000/gvfsd/socket-dq3j7IWG
unix    3      [ ]       STREAM    CONNECTED 10119
unix    3      [ ]       STREAM    CONNECTED 10854  /run/user/1000/at-spi-bus_0
unix    3      [ ]       STREAM    CONNECTED 11577
unix    3      [ ]       STREAM    CONNECTED 11571
unix    3      [ ]       STREAM    CONNECTED 8210  /run/systemd/journal/stdout
unix    2      [ ACC ]   STREAM    LISTENING 1667  /run/ssh-unix-local/socket
unix    3      [ ]       STREAM    CONNECTED 12374
unix    3      [ ]       STREAM    CONNECTED 9151  /run/user/1000/bus
unix    2      [ ACC ]   STREAM    LISTENING 1669  /run/systemd/io.systemd.Hostname
unix    3      [ ]       STREAM    CONNECTED 11761
unix    3      [ ]       DGRAM     CONNECTED 2787
unix    3      [ ]       STREAM    CONNECTED 11882
unix    3      [ ]       STREAM    CONNECTED 10110  /run/dbus/system_bus_socket
unix    3      [ ]       STREAM    CONNECTED 11576
unix    3      [ ]       STREAM    CONNECTED 12260  @/tmp/.X11-unix/X0
unix    3      [ ]       STREAM    CONNECTED 9126
```

Figure 5: System Process Monitoring in Kali Linux

Explanation:

Volatile data collection activities were performed to understand how SOC analysts and forensic investigators collect active system information during incident response operations. Network connection details, listening ports, and active communication sessions were documented as part of the evidence preservation process. This activity



helped improve understanding of live system analysis and volatile evidence collection procedures commonly followed during cybersecurity investigations.

Observation:

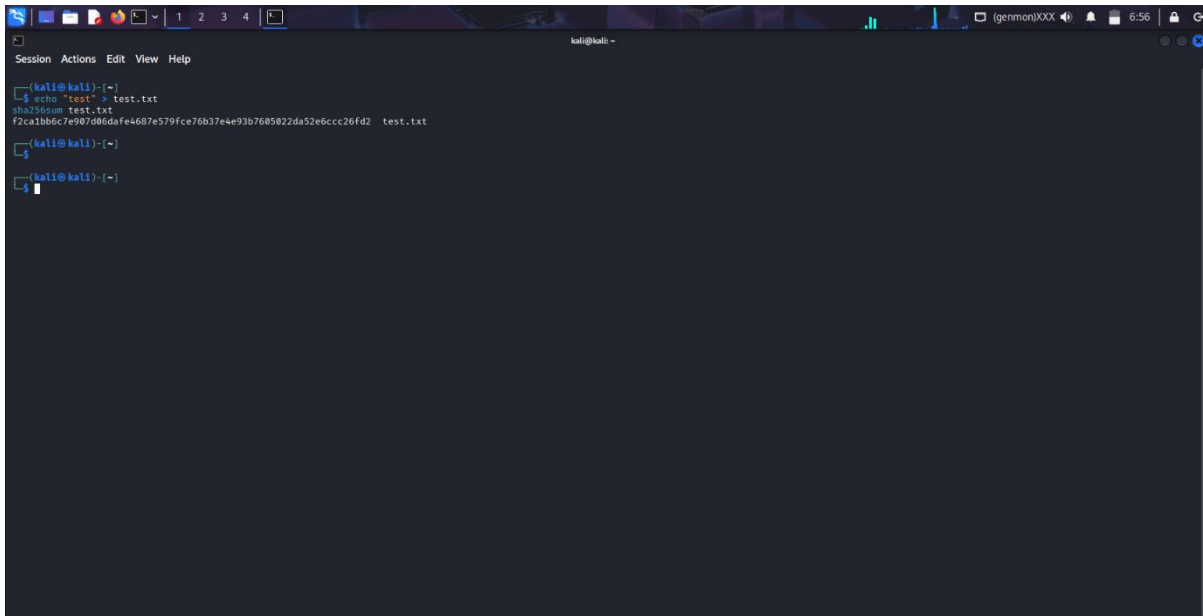
The volatile data collection process provided visibility into active system connections and running network sessions during monitoring operations. Live system analysis improved the ability to identify suspicious communication activity and potential unauthorized connections.

Learning Outcome:

This activity improved understanding of volatile evidence collection techniques, live system monitoring, and forensic investigation procedures commonly followed during SOC incident response operations.

Evidence Collection Documentation:

Evidence Item	Description	Collected By	Collection Date	Hash Value
Memory Dump	Server-Y Memory Acquisition	SOC Analyst	2026-05-20	SHA256:4F7A9B2C
Network Logs	Suspicious Traffic Logs	SOC Analyst	2026-05-20	SHA256:A91B72D4
Authentication Logs	Failed Login Attempts	SOC Analyst	2026-05-20	SHA256:C37D18E5



```
kali@kali: ~  
$ echo "test" > test.txt  
$ cat test.txt  
2ca1bb6c7e987d86d4fe4687e579fce76b37e4e93b7605822da52e6ccc26fd2 test.txt  
$  
$  
$
```

Figure 6: SHA256 Hash Verification for Evidence Integrity

Explanation:

Evidence collection procedures were documented to understand how digital evidence is preserved during cybersecurity investigations and incident response activities. Different evidence sources such as memory dumps, network logs, and authentication records were collected and documented along with SHA256 hash values to maintain integrity and support chain-of-custody requirements during forensic analysis operations.

Observation:

The evidence preservation process improved understanding of digital forensic documentation and integrity verification procedures. Hash values provided assurance that the collected evidence remained unaltered during investigation activities.

Learning Outcome:

This activity improved understanding of evidence preservation procedures, forensic documentation, hash verification techniques, and chain-of-custody practices commonly followed during cybersecurity investigations.



Chain of Custody:

The collected digital evidence was documented and preserved securely to maintain data integrity during the investigation process. Proper chain-of-custody procedures were followed to ensure that the evidence remained valid, traceable, and unaltered throughout forensic analysis and incident response activities.

Explanation:

Chain-of-custody procedures were documented to understand how digital evidence is securely handled and transferred during forensic investigations. Proper evidence management practices help maintain the authenticity and integrity of collected information throughout the investigation lifecycle.

Observation:

The chain-of-custody process improved understanding of secure evidence handling procedures and forensic documentation practices followed during cybersecurity investigations and incident response operations.

Learning Outcome:

This activity improved understanding of evidence integrity protection, secure forensic handling procedures, and chain-of-custody documentation practices commonly implemented during digital investigations.



Capstone Project: Full SOC Workflow Simulation

Attack Simulation:

Attack Scenario:

A simulated exploitation attempt targeting a vulnerable Samba service was performed to understand how SOC teams detect and respond to remote exploitation activities during security monitoring operations.

Tool Used:

Metasploit Framework

Target System:

Metasploitable2 Virtual Machine

MITRE ATT&CK Technique:

T1210 – Exploitation of Remote Services

```
kali@kali: ~  
$ msfconsole  
Metasploit tip: Use sessions -i to interact with the last opened session  
  
[*] Metasploit v6.4.126-dev  
+ --=[ 2,638 exploits - 1,331 auxiliary - 2,153 payloads ]  
+ --=[ 432 post - 49 encoders - 14 nops - 12 evasion ]  
Metasploit Documentation: https://docs.metasploit.com/  
The Metasploit Framework is a Rapid7 Open Source Project  
msf >
```

Figure 7: Metasploit Framework Initialization for Attack Simulation

Explanation:

The attack simulation activity was performed to understand how SOC analysts detect and investigate exploitation attempts targeting vulnerable systems. A simulated Samba exploitation scenario was documented to



demonstrate how suspicious activities are identified during monitoring operations. This activity helped improve understanding of attack detection workflows, exploitation monitoring procedures, and SOC investigation processes commonly followed during incident response operations.

Observation:

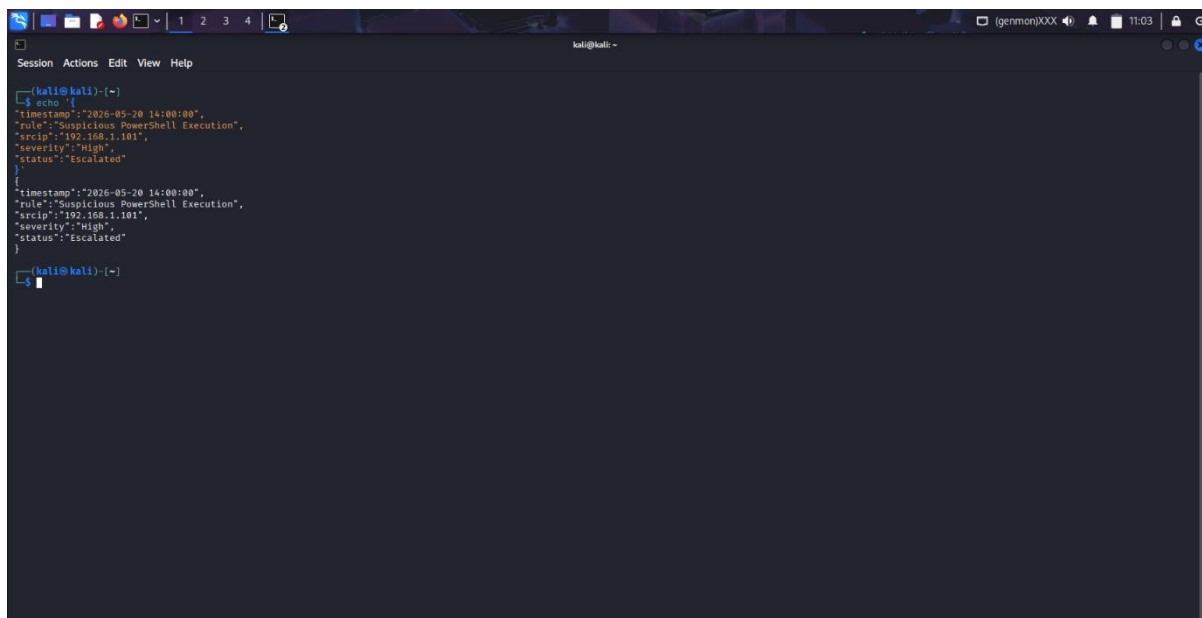
The simulated attack activity generated suspicious network behavior and exploitation-related indicators during monitoring operations. The attack simulation improved visibility into exploitation attempts targeting vulnerable services within monitored environments.

Learning Outcome:

This activity improved understanding of attack simulation concepts, exploitation monitoring techniques, and SOC investigation workflows commonly followed during cybersecurity incident response activities.

Detection and Triage:

Timestamps	Source IP	Alert Description	MITRE Technique	Status
2026-05-20 14:00:00	192.168.1.101	Samba Exploit Attempt	T1210	Escalated
2026-05-20 14:05:00	192.168.1.102	Suspicious PowerShell Activity	T1059	Investigating
2026-05-20 14:10:00	192.168.1.103	Abnormal Authentication Attempt	T1078	Open



```
(kali@kali)~$ echo '{
  "timestamp": "2026-05-20 14:00:00",
  "rule": "Suspicious PowerShell Execution",
  "srcip": "192.168.1.101",
  "severity": "High",
  "status": "Escalated"
}'
{
  "timestamp": "2026-05-20 14:00:00",
  "rule": "Suspicious PowerShell Execution",
  "srcip": "192.168.1.101",
  "severity": "High",
  "status": "Escalated"
}
(kali@kali)~$
```

Figure 8: Simulated Wazuh Security Alert Monitoring

Explanation:

The detection and triage process was performed to identify and prioritize suspicious exploitation activities generated during the attack simulation scenario. Different alerts such as Samba exploitation attempts, suspicious PowerShell activity, and abnormal authentication behavior were analyzed using timestamps, source IP addresses, and MITRE ATT&CK techniques. This activity improved understanding of SOC alert investigation and prioritization workflows.

Observation:

The generated alerts provided visibility into suspicious exploitation attempts and abnormal system activities during monitoring operations. Alert triage procedures improved incident prioritization and supported additional investigation activities.

Learning Outcome:

This activity improved understanding of alert detection workflows, incident prioritization procedures, and SOC triage techniques commonly followed during cybersecurity monitoring operations.



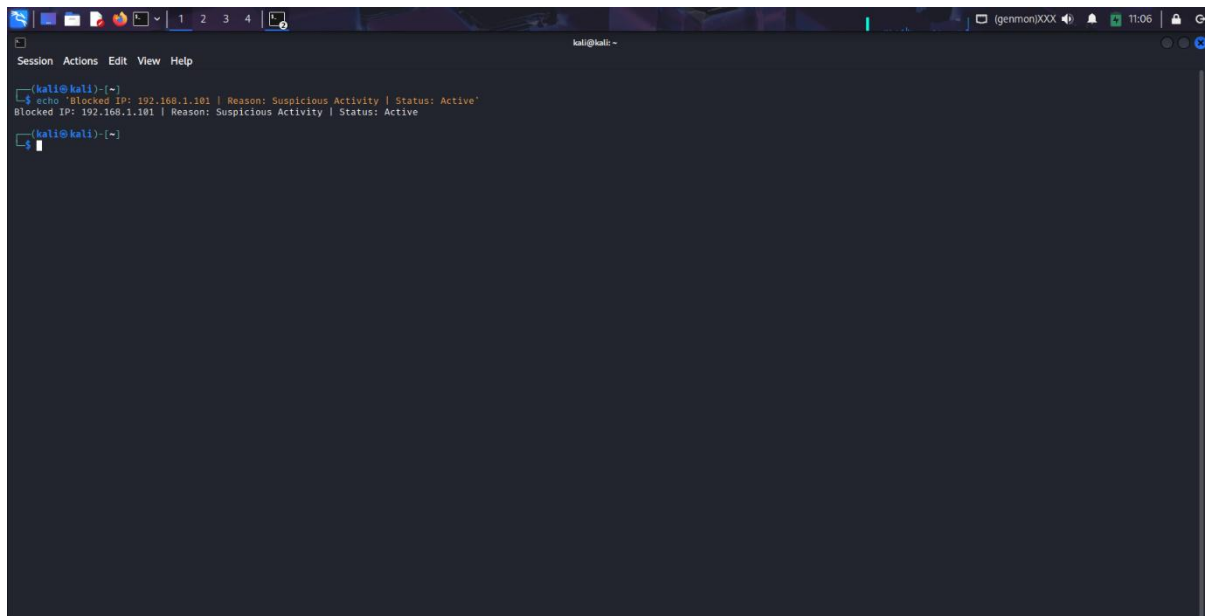
Response and Containment

Response Actions:

- Isolated the affected virtual machine
- Blocked suspicious source IP addresses
- Monitored abnormal network traffic
- Escalated incident to Tier 2 analysts
- Documented investigation findings

Containment Status:

Successful



```
kali@kali: ~  
$ echo 'Blocked IP: 192.168.1.101 | Reason: Suspicious Activity | Status: Active'  
Blocked IP: 192.168.1.101 | Reason: Suspicious Activity | Status: Active  
$
```

Figure 9: CrowdSec Blocked IP Monitoring

Explanation:

Response and containment activities were performed to reduce the impact of the simulated exploitation attempt during monitoring operations. The affected system was isolated and suspicious network activity was monitored to prevent additional exploitation attempts. This activity improved understanding of incident response coordination and containment procedures commonly followed during SOC operations.



Observation:

The response workflow improved visibility into containment procedures and incident handling coordination during security monitoring activities. Isolation and monitoring procedures reduced the impact of suspicious activity within the simulated environment.

Learning Outcome:

This activity improved understanding of incident containment procedures, response coordination, and SOC remediation workflows commonly implemented during cybersecurity investigations.

Incident Report:

Executive Summary:

A simulated exploitation attempt targeting a vulnerable Samba service was detected during SOC monitoring operations. Suspicious exploitation activity and abnormal authentication behavior were identified and analyzed during the investigation process. The affected system was isolated successfully and the incident was escalated for advanced investigation.

Timeline:

- Attack detected at 14:00
- Alert triage completed at 14:10
- Containment actions initiated at 14:20
- Incident escalated to Tier 2 at 14:30

Recommendations:

- Apply security patches regularly
- Restrict unnecessary network services
- Monitor suspicious authentication activity
- Improve threat monitoring and alerting procedures



Explanation:

The incident report was prepared using a structured SANS-style reporting format to document the attack scenario, investigation timeline, response actions, and security recommendations. Structured reporting procedures improve incident documentation quality and support effective communication during cybersecurity investigations.

Observation:

The structured reporting process improved visibility into investigation timelines, incident response activities, and security recommendations generated during the SOC workflow simulation exercise.

Learning Outcome:

This activity improved understanding of incident reporting procedures, SANS-style documentation practices, and structured SOC communication workflows commonly followed during cybersecurity investigations.

Management Briefing:

A suspicious exploitation attempt targeting a vulnerable system was detected and analyzed during SOC monitoring operations. The affected system was isolated successfully, and response procedures were initiated to reduce the impact of the incident. The investigation findings were documented and escalated for advanced analysis while continuous monitoring activities were maintained throughout the incident response process.

Explanation:

The management briefing was prepared to summarize the incident scenario, response activities, and investigation outcomes in a simplified and structured format suitable for non-technical stakeholders. Effective communication procedures improve incident awareness and support decision-making during cybersecurity response operations.



Observation:

The management briefing process improved understanding of structured communication procedures and simplified incident reporting techniques used during SOC operations and cybersecurity investigations.

Learning Outcome:

This activity improved understanding of management communication procedures, simplified incident reporting practices, and SOC stakeholder briefing workflows commonly followed during cybersecurity incidents.

Final Observation:

The overall practical activities provided detailed understanding of advanced SOC operations, threat intelligence integration, incident escalation workflows, alert triage procedures, evidence preservation techniques, and full SOC workflow simulation processes. Different monitoring, investigation, escalation, and reporting activities improved understanding of how SOC analysts detect, investigate, prioritize, and respond to cybersecurity incidents within monitored environments.

Conclusion:

This practical task successfully demonstrated advanced SOC monitoring and incident response concepts through different workflow simulations and documentation activities. The performed tasks improved understanding of log analysis, threat intelligence usage, alert triage, evidence preservation, incident escalation, and attack response procedures commonly followed in SOC environments. The capstone project further improved practical understanding of complete SOC investigation workflows and cybersecurity incident response operations.